

WORK IN PROGRESS

Enterprise Retail & Strategic Inventory System

Group 5: Invo6

BSc (Hons) Artificial Intelligence, Level 5

System Development Group Project

The British College, Kathmandu

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1. Introduction

Invo6 is a web and mobile-based Enterprise Retail and Strategic Inventory System we are building as our System Development group project. The goal is to give retail shop owners in Nepal a single platform to manage billing, inventory, and sales analytics, replacing handwritten billing with a fast, barcode-driven system that also gives customers access to their own purchase history through a mobile app.

This report covers the work completed up to the Work-In-Progress submission. It explains the current status of the project, the progress made in each phase, screenshots of what is built so far, and the solutions adopted when we ran into problems.

2. Current Status of the Project

At the time of this WIP submission, the project is on track. The Initiation and Planning phases are fully complete. We are well into the Executing phase, with all design artefacts submitted and active development underway on both the web dashboard and the mobile app.

On the development side, the backend REST API is in progress using FastAPI and MySQL. The React-based web dashboard has its initial layout and authentication flow being built.

3. Progress Made in Each Phase

3.1. Initiation Phase

This phase was about forming a team, agreeing on the problem to solve, and producing the foundation documents that would guide the rest of the project. We identified the pain points of manual billing in Nepalese retail shops and scoped a system that would be affordable and practical for independent stores.

Deliverables completed in this phase:

- **Project Title document**
- **Project Charter**
- **Team Contract**
- **Risk Register**

3.2. Planning Phase

In this phase, we focused on understanding the team's strengths, setting up collaboration tools, and designing the workflow we would follow during development.

Deliverables completed in this phase:

- **Belbin Analysis:** Each member completed the test, and roles are chosen based on natural team strengths.
- **Skill Audit:** Individual and combined team audits are used to identify gaps and decide who handles what area.
- **Communication Tool:** Discord for regular meetings, GitHub for version control and code collaboration, Email for formal communication, and Google Drive for file storage, sharing, and collaboration document management.
- **Project Management (Jira):** Workspace created and initial setup done, team is beginning to use it for task tracking as development ramps up.

3.3. Execution Phase

This is the phase we are currently in, and it carries the most weight. It is split into two parts: design artefacts and product development.

Design Artefacts (complete):

- **Use Case Diagrams:** Separate diagrams for Customer, Cashier, and Admin roles, plus a combined overview diagram.
- **ERD:** Basic ERD, EERD, final composite/de-composite ERD, and logical table.
- **Wireframes:** Full wireframe set for both the React Native mobile app and the React web dashboard, covering all major screens.
- **Logo:** Two logos are designed for Invo6.
- **Requirement Catalogue:** The functional requirements include product and inventory management, customer features, shopkeeper/admin features, billing and transactions, AI and analytics, IoT integration, authentication and security, and store administration. The non-functional requirements define the system's performance, usability, reliability, and security standards.

Product Development (in progress):

We are following Agile Scrum tracked in Jira. Development is ongoing across the following areas:

- **Backend (FastAPI + MySQL):** core models defined, REST API endpoints for products, inventory, billing sessions, transactions, and user management in progress. JWT auth, bcrypt hashing, and email OTP 2FA implemented.
- **IoT Integration (ESP32 + MQTT):** barcode scanner module integrated, ESP32 publishes scan events to the backend via MQTT, which updates the active billing session in real time via WebSocket.
- **Web Dashboard (React):** authentication flow (login, register, 2FA, forgot password) and main dashboard layout built.
- **Mobile App (React Native):** onboarding carousel, authentication screens, and customer transaction history dashboard built and connected to the backend.

- **AI Module:** sales forecasting regression model and RAG-based chatbot (using FAISS vector database and sentence embeddings) in development.
- **Docker Compose:** containerization configuration in progress for one-command deployment.

Project Management:

We have set up Jira as our project management tool and are in the early stages of using it. The workspace has been created, and we are beginning to log tasks as development ramps up. Alongside Jira, we have held 12 documented team meetings on Discord and in class to coordinate work, review submissions, and align on next steps. GitHub is used for version control with a clear branching strategy (main/develop/feature/*).

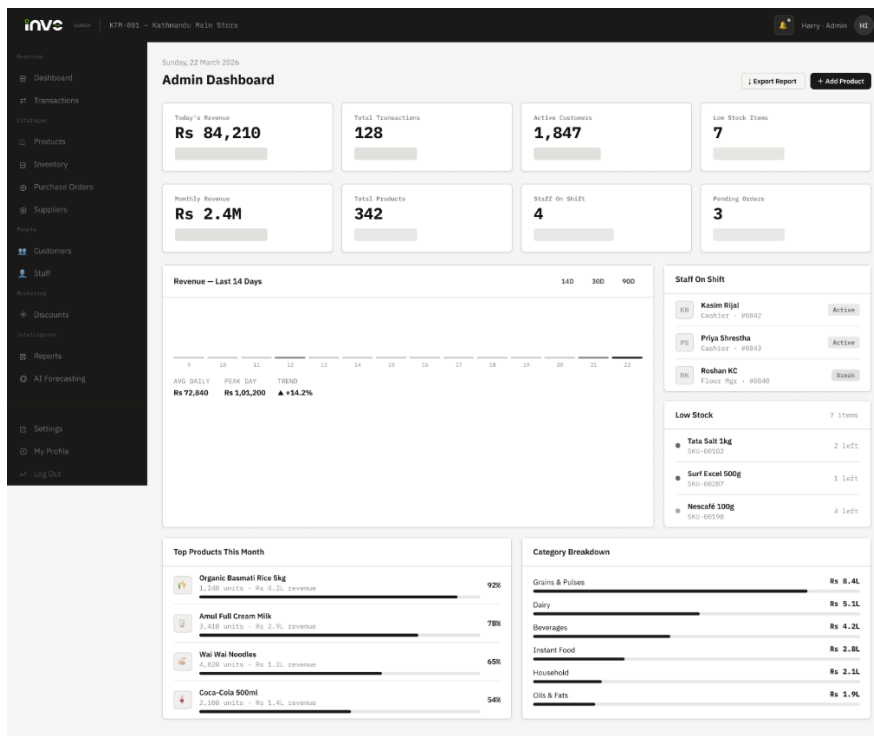
4. Relevant Screenshots

The original files are in the Executing folder of this submission.

4.1. Wireframe

Only some wireframe screenshots are shown here. The rest can be found in the wireframe folder.

Web Dashboard for Admin:



Admin Dashboard Page

inve

KTM-001 - Kathmandu Main Store

Harry - Admin

HE

Dashboard

Transactions

Products

Inventory

Purchase Orders

Suppliers

Customers

Staff

Discounts

Reports

All Forecasting

Settings

My Profile

Log Out

Real-time stock levels

Inventory

Total Stock

342

In Stock

318

Low Stock

7

Out of Stock

17

Filter

Product	SKU	Category	Stock	Reorder At	Last Restock	Supplier	Status	
Organic Basmati Rice 5kg	SKU-00412	Grains	148	20	15 Mar	Nepal Trading	In Stock	Adjust
Amul Full Cream Milk 1L	SKU-00218	Dairy	240	50	20 Mar	Himalaya Dist.	In Stock	Adjust
Tata Salt 5kg	SKU-00102	Condiments	2	15	1 Mar	Himalaya Dist.	Low Stock	Reorder
Surf Excel Detergent 500g	SKU-00287	Household	0	10	8 Feb	ClearSupply	Out of Stock	Reorder
Nescafé Classic 100g	SKU-00206	Beverages	4	20	5 Mar	Himalaya Dist.	Low Stock	Reorder

Showing 1-5 of 342 items

1

2

Admin Inventory Page

inve

six

YOUR TRUSTED BILLING PARTNER

Welcome back

Sign In

Sign in as

Cashier

Admin

Email Address

kasim@store.np

Password

Remember this device

Forgot password?

Sign In to Terminal

2FA VERIFICATION

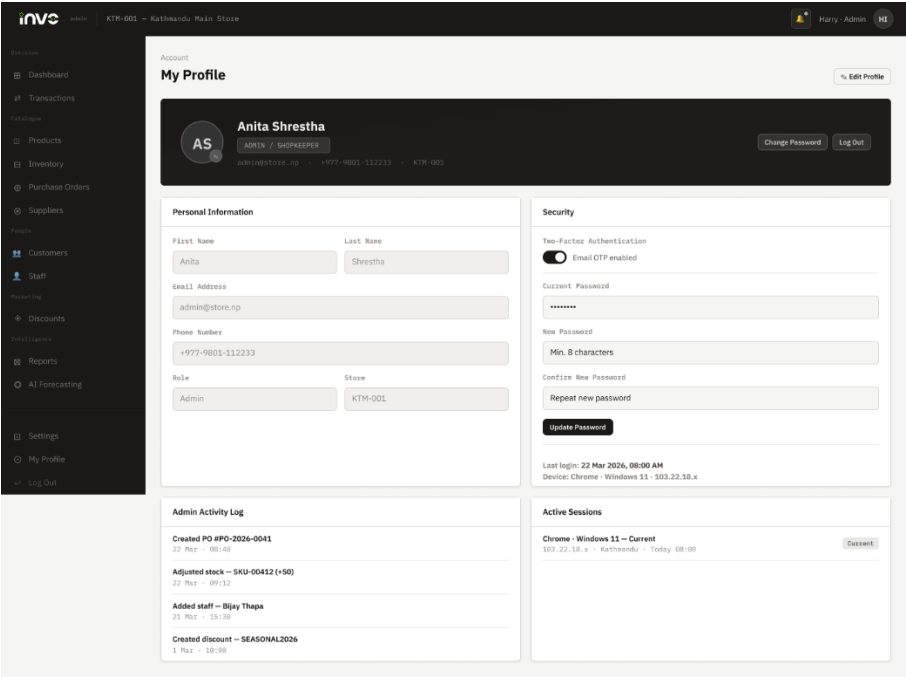
Email OTP Required

A 6-digit code will be sent to your registered email address before terminal access is granted.

Admin Login page

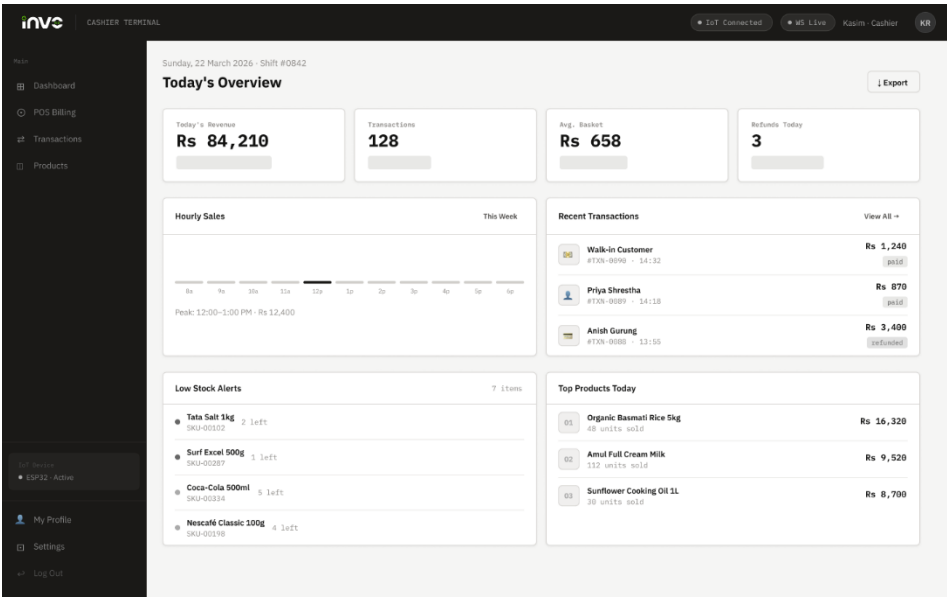


OTP Verification Page

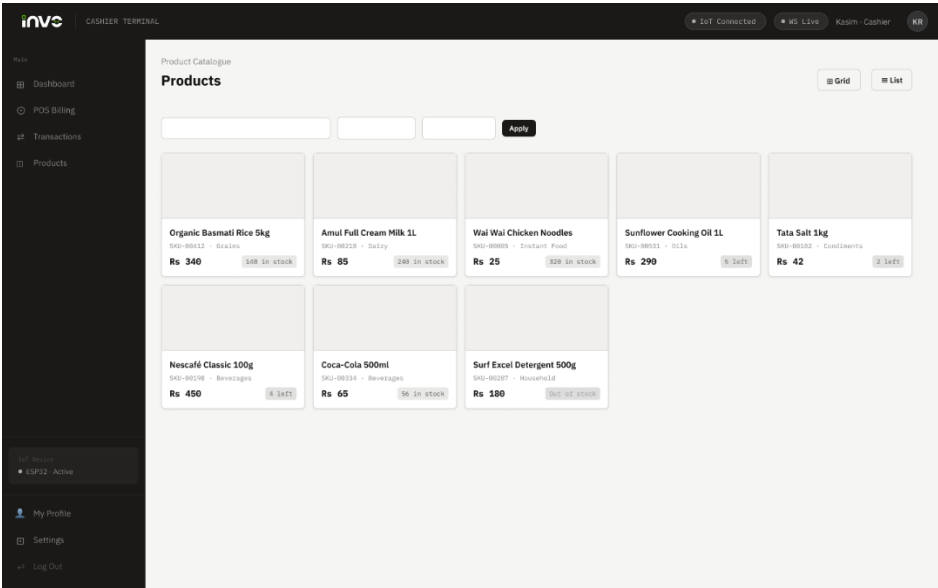


Admin Profile Page

Web Dashboard for Cashier:



Cashier Dashboard Page



Cashier Products Page

STORE

New Baneshwor, Kathmandu
Tel: +977-1-441-0000
PAN: PAN-00112233

TXN: #TXN-20260322-0091

Date: 22 Mar 2026 Time: 14:48

Cashier: Kasim R. Shift: #0842

Customer: Rohan Sharma
Phone: +977-9841-234567

ITEM	AMT
Basmati Rice 5kg	
2 × Rs 340.00	Rs 680.00
Amul Milk 1L	
3 × Rs 85.00	Rs 255.00
Wai Wai Noodles	
6 × Rs 25.00	Rs 150.00
Sunflower Oil 1L	
1 × Rs 290.00	Rs 290.00

Subtotal Rs 1,375.00

Discount (10%) - Rs 137.50

VAT (13%) Rs 178.75

TOTAL Rs 1,416.25

Method: Cash

Tendered: Rs 1,500.00

Change: Rs 83.75

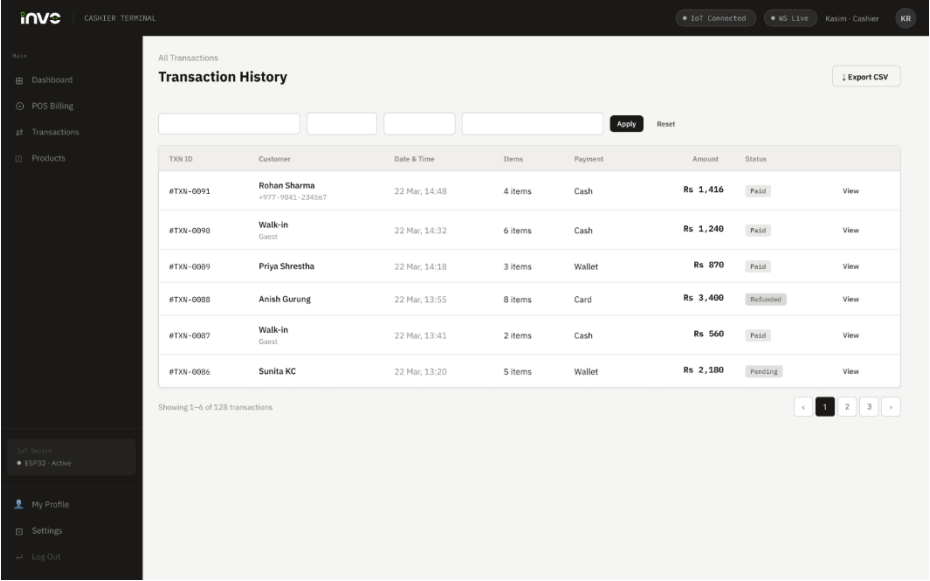


Thank you for shopping with us!
Exchange within 7 days with receipt.
www.invo6store.np

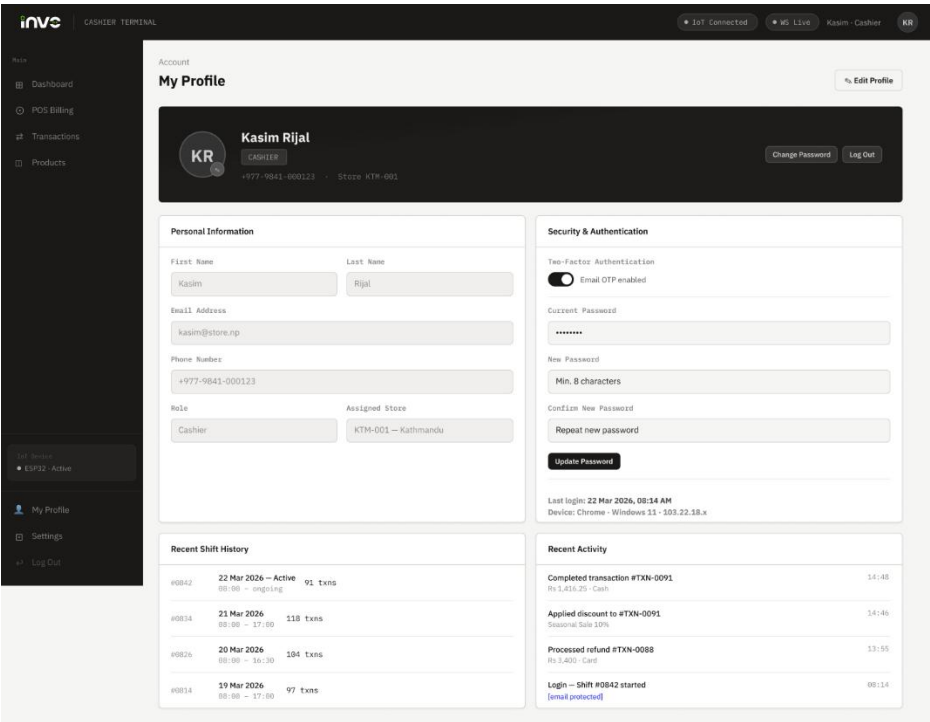


TXN20260322 0091

Receipt Preview

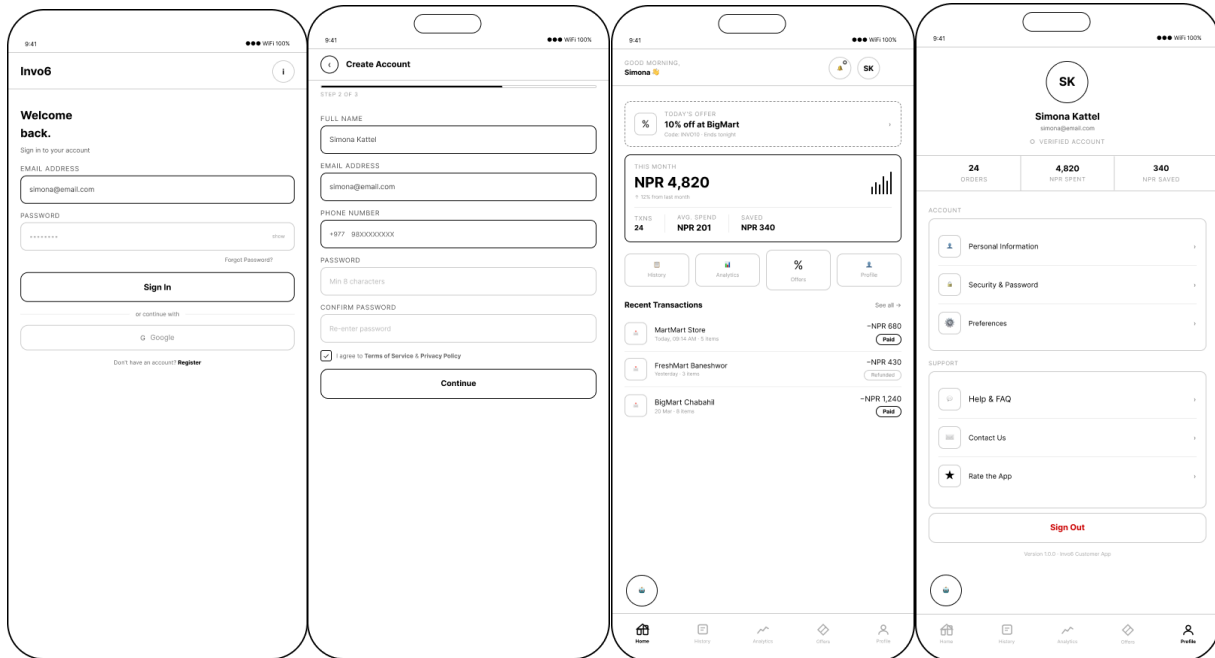


Transaction History Page



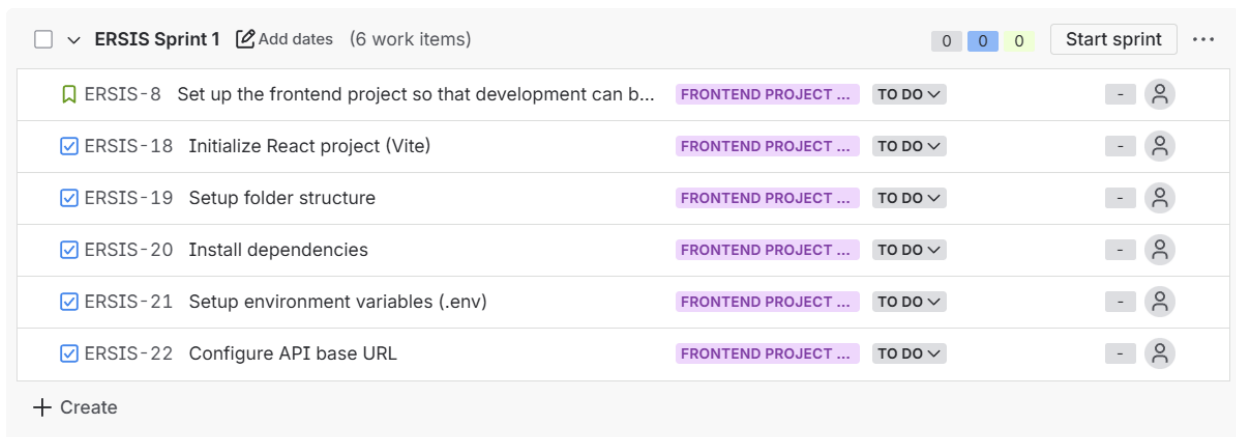
Cashier Profile Page

Mobile App for Customer:



4.2. Project Management (Jira)

Jira has been set up as our project management tool. We are in the early stages of populating it with tasks. Below is a screenshot of the current workspace setup.



Backlog

(36 work items)

0

0

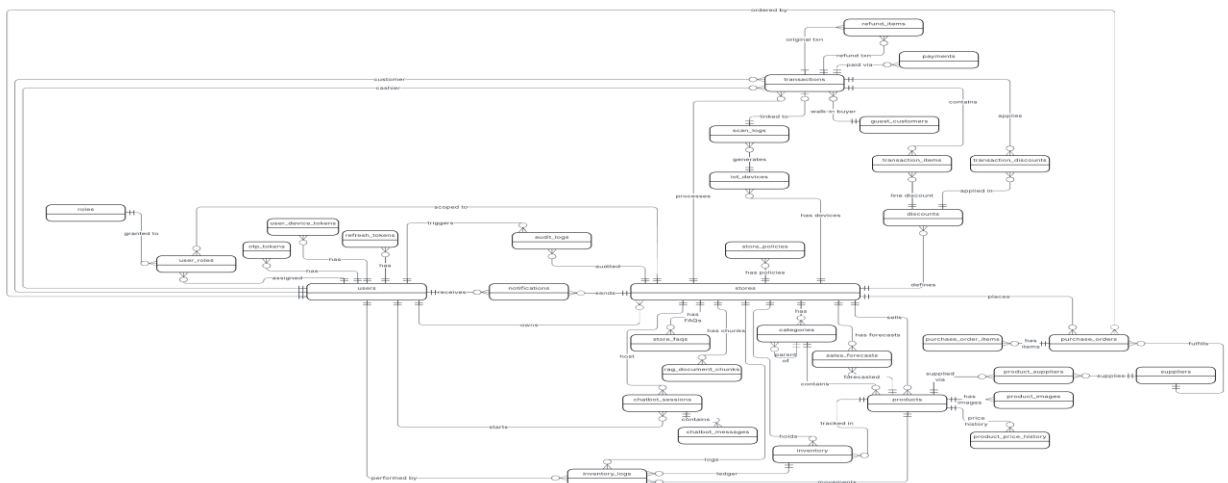
0

Create sprint

ERSIS-9	As a user, I want to login/register so that I can acces...	AUTHENTICATION UI	TO DO	-	
ERSIS-10	As an admin, I want to manage products so that inve...	ADMIN DASHBOARD	TO DO	-	
ERSIS-11	As an admin, I want to view sales data so that I can ...	ADMIN DASHBOARD	TO DO	-	
ERSIS-12	As a cashier, I want to create bills quickly so that cu...	CASHIER DASHBOAR...	TO DO	-	
ERSIS-13	As a cashier, I want barcode scanning input so that ...	CASHIER DASHBOAR...	TO DO	-	
ERSIS-14	As a customer, I want to view my purchase history s...	CUSTOMER DASHBOA...	TO DO	-	
ERSIS-15	As a customer, I want to see my profile so that I can...	CUSTOMER DASHBOA...	TO DO	-	
ERSIS-16	As a user, I want real-time updates so that data is all...	REAL-TIME FEATURES	TO DO	-	
ERSIS-17	As a user, I want a clean UI so that the system is ea...	UI/UX ENHANCEMENTS	TO DO	-	
ERSIS-23	Create Login Page UI	AUTHENTICATION UI	TO DO	-	
ERSIS-24	Create Registration Page UI	AUTHENTICATION UI	TO DO	-	
ERSIS-25	Form validation (email, password rules)	AUTHENTICATION UI	TO DO	-	

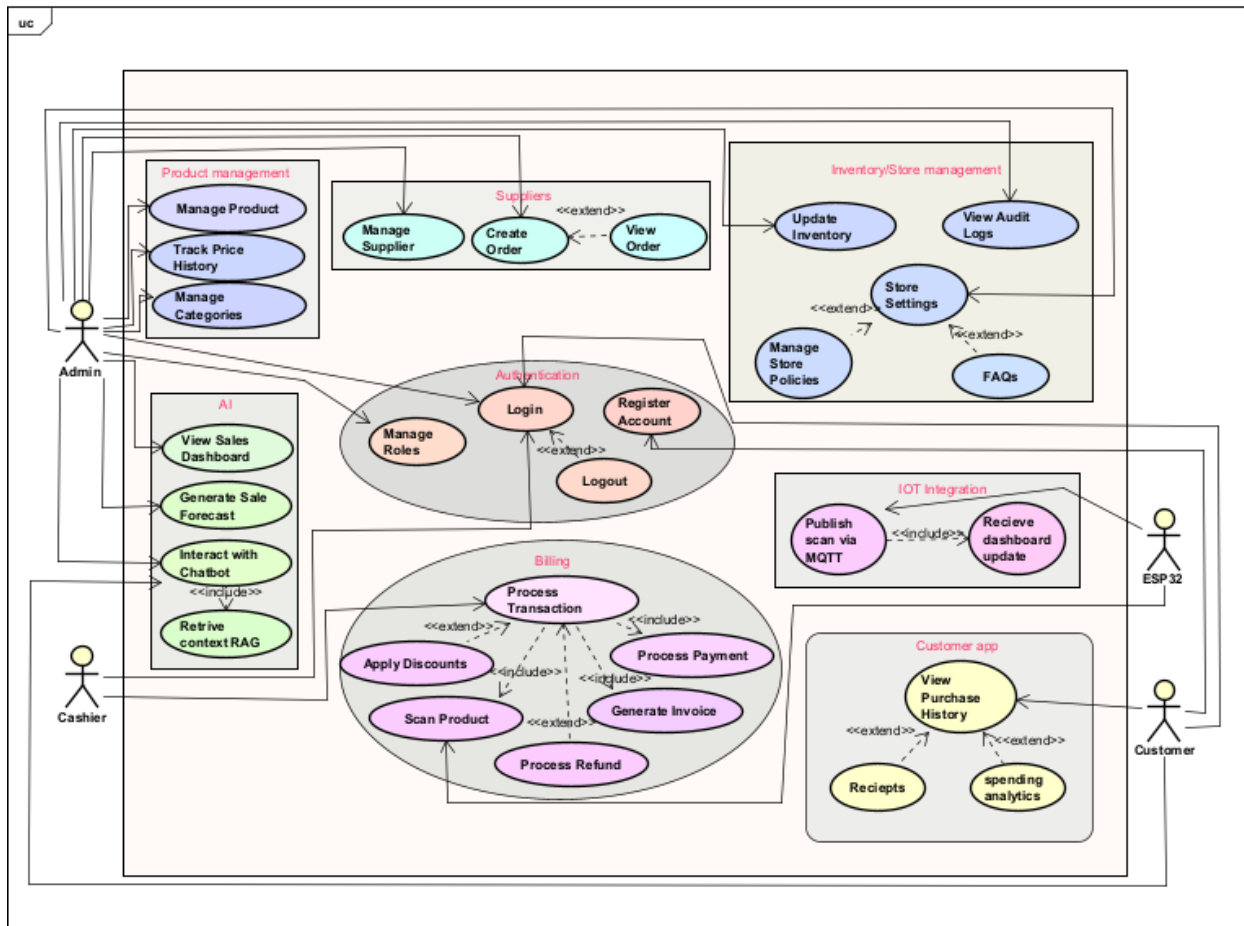
4.3. Design Artefacts:

Entity Relationship Diagrams:



Entity Relationship Diagram

4.4. Use Case Diagrams



Overview (All Actors)

4.5. Logo



six

Logo 1



six

Logo 2

5. Team Meetings

Regular team meetings were conducted to ensure alignment, track progress, and address challenges throughout the project.

Meeting 1

- **Date:** 21/02/2026
- **Objective:** Initial team introduction and project discussion.
- **Discussion Points:**
 - **Introduction of all team members**
 - **Overview of the project**
 - **Discussion on individual contributions and roles**
- **Meeting Image:**

Meeting 2

- **Date:** 22/02/2026
- **Objective:** Quick project alignment
- **Discussion Points:**
 - Short discussion on project direction
 - Clarification of initial ideas

Meeting 3

- **Date:** 24/02/2026
- **Objective:** Project planning and documentation
- **Discussion Points:**
 - Team contract submission
 - Project charter discussion

Meeting 4

- **Date:** 11/03/2026
- **Objective:** Team structure and idea development
- **Discussion Points:**
 - Skill audit discussion
 - Team contract review
 - Logo design ideas
 - Belbin analysis
 - Overall project concepts

Meeting 5

- **Date:** 14/03/2026
- **Objective:** Team identity and role assignment
- **Discussion Points:**
 - Finalization of group name
 - Skill audit discussion
 - Assignment of roles and responsibilities

Meeting 6

- **Date:** 18/03/2026
- **Objective:** Submission review
- **Discussion Points:**
 - Belbin analysis submission

Meeting 7

- **Date:** 20/03/2026
- **Objective:** Design and submission planning
- **Discussion Points:**
 - **Submission requirements**
 - **Wireframe discussion**

Meeting 8

- **Date:** 21/03/2026
- **Objective:** Design refinement
- **Discussion Points:**
 - Continued wireframe discussions
 - Feedback and improvements

Meeting 9

- **Date:** 2/04/2026

- **Objective:** Development progress discussion
- **Discussion Points:**
 - Git clone process discussion
 - Work-in-progress submission updates
 - Coordination on ongoing development tasks

Meeting 10

- **Date:** 04/04/2026
- **Objective:** Review of project progress and version control coordination
- **Discussion Points:**
 - Working progress updates from team members
 - GitHub usage and collaboration discussion
 - Coordination on code integration and repository management

Meeting 11

- **Date:** 05/04/2026
- **Objective:** Submission of project requirements
- **Discussion Points:**
 - Requirement catalogue submission
 - Final review of documented requirements
 - Clarifications and feedback on submission

Meeting 12

- **Date:** 09/04/2026
- **Objective:** Review of ongoing development progress
- **Discussion Points:**
 - Updates on current working progress
 - Discussion of completed and pending tasks
 - Coordination on next steps and task distribution

6. Conclusion

The Invo6 system is progressing as planned, with all initiation and planning activities completed and development actively underway. Key design artefacts have been finalized, and core system components, including backend, web dashboard, mobile app, and IoT integration, are in development. The team is effectively collaborating using Agile practices and project management tools. Overall, the project is on track to deliver a practical, efficient, and scalable retail and inventory solution for Nepalese shop owners.